MILITARY SPECIFICATION

CABINET, STORAGE, MILK SHAKE, MECHANICALLY REFRIGERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an upright, self-contained, mechanically refrigerated, milk shake storage cabinet.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants

MILITARY

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment Communications, Electronic and Associated Electrical Equipment)

MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement Of

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A 276 - Stainless and Heat-Resisting Steel Bars and Shapes
A 463 - Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1
A 582 - Free Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished
B 280 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service
C 177 - Steady State Thermal Transmission Properties by Means of the Guarded Hot Plate

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 7 - Food Service Refrigerators and Storage Freezers

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 471 - Commercial Refrigerators

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The milk shake storage cabinet shall be the manufacturers current standard product, except for the changes necessary to comply with the requirements of this specification.

3.2 Codes and standards. The milk shake storage cabinet shall comply with the applicable requirements of NSF 7 and UL No. 471.

3.2.1 Compliance. Prior to the approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the milk shake storage cabinet he proposes to furnish under this specification meets the applicable requirements of NSF and UL.

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished milk shake storage cabinet.

(2) A certification for the milk shake storage cabinet issued under their special one-time contract evaluation/certification service.

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(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the milk shake storage cabinets are being procured, indicating that the milk shake storage cabinets have been tested and conform to NSF.

3.2.1.2 UL. Acceptable evidence of meeting the applicable requirements of UL shall be the UL Listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the milk shake storage cabinet has been tested and conforms to UL Standard No. 471.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer provided the completed items comply with all the provisions of this specification (see 6.3).

3.3.1 Refrigeration tube. Copper tube refrigeration lines shall conform to ASTM B 280. Tube used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness of not less than 0.016 inch.

3.3.2 Bolts, nuts, screws, and washers. Bolts, nuts, screws, and washers used for mounting components shall be of the quality and in the sizes, types, and materials normally used for the purpose in commercial practice. Threads shall conform to FED-STD-H28. Steel bolts, nuts, screws, and washers for the evaporator shall be zinc plated. Brass bolts, nuts, washers, and machine screws shall not be used in fastening aluminum parts.

3.3.3 Insulation. All insulation shall be expanded polyurethane and shall have a "K" factor, based on ASTM C 177 determined by using a 75°F mean temperature, expressed in British thermal units per hour (Btu/hr) per square foot per inch of thickness per degree F temperature difference. Loose fill type insulation shall not be acceptable.

3.3.3.1 Foamed-in-place. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial "K" factor of not more than 0.15 and a density of not less than 1.9 pounds per cubic foot when tested as specified in 4.4.7.

3.3.4 Stainless steel plates, sheets, and strips. The stainless steel plates, sheets, and strips shall conform to any type 300 series of ASTM A 167.

3.3.5 Stainless and heat-resisting steel bars. Stainless and heat-resisting steel bars shall conform to ASTM A 276 or ASTM A 582.

3.3.6 Aluminum coated steel. Aluminum coated steel shall conform to ASTM A 463.

3.3.7 Refrigerant. The refrigerant shall conform to type 12 of BB-F-1421.

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3.4 Design and construction. The milk shake storage cabinet shall be designed to maintain 8 full baskets of milk shakes at 18°F when operating in an ambient temperature of 90°F. The milk shake storage cabinet shall be of the upright design equipped with a self-contained refrigeration system and controls necessary for automatic operation. Hereinafter, the milk shake storage cabinet shall be referred to as the shake cabinet. The unit shall be ready for operation after removal of shipping protection, opening of valves and connection to electric power. All accessories, instruments, controls and tubing shall be fastened or braced. The shake cabinet overall dimensions shall be 30 inches wide by 35 inches deep by 78 inches high maximum with a dimensional tolerance of ± 1/2 inch.

3.4.1 Cabinet. The cabinet shall form a completely rigid structure. The cabinet shall consist of an outer shell, refrigerated compartment liner, base, door, door hinges, latch and striker assembly and lock assembly. The cabinet and condensing unit shall have an integral structural frame. The structural frame shall be of the full perimeter design, constructed of welded members. The space between the outer shell and refrigerated compartment liner shall be completely filled with a minimum of 2 inches of insulation specified in 3.3.3.1. Electric anticondensate heaters shall be installed in the doors, door mullions, and sills. Tapping plates for all hardware shall be installed when machine screws are used. Tapping plates shall have a contact surface with the refrigerated compartment liner or outer shell at least equal to 125 percent of the contact surfaces of the item being fastened and the threaded screw holes shall be not closer than 1/2 inch. Self tapping screws are not allowed. The base exterior shall be fabricated from aluminum coated steel specified in 3.3.6. The shake cabinet insulated areas shall have no visible condensation on the outside of the cabinet or doors when tested as specified in 4.4.3.2.

3.4.1.1 Outer shell, top and rear. The outer shell shall be fabricated from stainless steel specified in 3.3.4 having a minimum thickness of 0.0375 inch. The top and rear shall be fabricated from galvanized steel or stainless. Joints shall be spot or seam welded where practicable. Joints that are not welded shall be sealed to prevent moisture infiltration into the insulation. The outer shell shall totally enclose the condensing unit and shall incorporate a louver in the top front of the cabinet for ventilating the condensing unit equipment.

3.4.1.2 Refrigerated compartment liner. The refrigerated compartment liner shall be fabricated from stainless steel specified in 3.3.4 having a minimum thickness of 0.031 inch. Seams shall be either locked and soldered, or locked, spot welded, and sealed to prevent moisture infiltration into the insulation and form a permanent water tight seal except for any openings in the liner designed for vapor migration.

3.4.1.3 Breaker strips. Breaker strips shall be fabricated of thermoplastic material, thermosetting plastic, natural rubber, or synthetic rubber and shall be installed to join the outer shell and inner liner around the door opening. Joints in breaker strips shall be smooth and cleanable. Edges of the breaker strips shall be eased or rounded. The entire installation shall provide a smoothly finished appearance.

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3.4.2 Door. The cabinet shall be provided with one full length door, hinged right or left, as specified (see 6.2). The door shall be of the overlapping type. The door exterior shall be fabricated of stainless steel specified in 3.3.4 having a minimum thickness of 0.031 inch. The door shall have a welded frame with hinge and latch tapping plates integral with or welded to the frame. Door frames shall be equipped with breaker strips with heater wires for condensation control. The door space between the door exteriors shall be completely filled with insulation specified in 3.3.3.1. The door shall be provided with a magnetic gasket. Each door shall be equipped with means for holding it in the closed position with sufficient force to effect a tight gasket seal when tested as specified in 4.4.2. The door exterior shall be provided with a handle. The door miter joints shall be continuous welded and sealed.

3.4.2.1 Gasket. The door shall be provided with a replaceable push in magnetic gasket. The magnetic gasket shall be one piece, close fitting and symmetrically installed to provide air-tight sealing.

3.4.2.2 Door hardware. The cabinet shall be provided with door hardware consisting of two hinges, door latch and strike assembly, door handle and door cylinder lock and lock keeper.

3.4.2.2.1 Hinges. Hinges for the cabinet door shall match the latch and strike assembly in finish and appearance. The hinges shall be fabricated of stainless steel specified in 3.3.4. Hinges shall be of the gravity action type and shall remain fully open to allow removal of the storage baskets. The hinges shall have a 120 degree of dwell. Hinges shall be removable.

3.4.2.2.2 Latch and strike assembly. The door latch and strike assembly shall be of the commercial heavy duty quality. Door latch and strike assembly shall be fabricated of stainless steel specified in 3.3.4 and 3.3.5, as applicable.

3.4.2.2.3 Handle. The door handle shall be fabricated from stainless steel specified in 3.3.4 and 3.3.5, as applicable.

3.4.3 Supports. The cabinet shall be provided with eight sets of angle supports to accomodate eight milk shake baskets (see 3.4.9). The angle supports shall be a minimum of 1-3/8 inch by 1-3/8 inch minimum and located 7 inches on center. The angle supports shall be fabricated from a minimum 0.0625 inch thick stainless steel specified in 3.3.4. The angle supports shall be installed to allow the milk shake baskets to be held in a plane parallel to the floor of the refrigerated compartment without tipping or rocking. The angle supports shall be removable for cleaning without the use of tools. The angle supports shall be capable of withstanding a static load of 50 pounds on each shelf without any deformation or distortion when tested as specified in 4.4.9.

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3.4.4 Thermometer. A remote reading, flush mounted thermometer shall be installed on the front and near the top outside of the door body so as to be accessible for viewing at all times. The bulb shall be mounted to permit circulation of air around the bulb and where it shall be out of the direct discharge of air from the evaporator. The thermometer shall have a range to permit accurate temperature readings between 0°F to 60°F.

3.4.5 Legs. The cabinet shall be provided with four stainless steel legs. The legs shall be a minimum of 6 inches high. Each leg shall be provided with a bottom flange with clearance holes for hold-down bolts. Each hold-down bolt shall be accessible for mounting and dismounting the cabinet. The legs shall be located at or near each corner and shall be attached to the bottom of the cabinet by bolts through gusset plates to provide support to the bottom of the cabinet.

3.4.6 Refrigeration system. The refrigeration system shall be self-contained within the framework and shall be designed to meet the requirements of 3.4 using the refrigerant specified in 3.3.7. The refrigeration system shall consist of an electric motor driven refrigeration compressor, air-cooled condenser, controls, and all other accessories necessary to operate as a completely automatic refrigeration system. All refrigerant components shall be gas tight when tested as specified in 4.4.4.

3.4.6.1 Air-cooled condenser. The air-cooled condenser shall be of the forced air type, constructed of brass, copper, aluminum alloy, or steel suitable protected on the outside after fabrication to resist corrosion. Fins shall be soldered, welded, or mechanically bonded to the tubes to insure permanent contact and shall be spaced not more than eight per inch. For uniform air flow, a fan shroud shall be provided.

3.4.6.2 Dehydration and charging. The completely assembled refrigeration system shall be evacuated, dehydrated, and charged with the refrigerant specified in 3.3.7 and quantity of oil recommended by the manufacturer.

3.4.6.3 Control system. The control system shall automatically start and stop the condensing unit to maintain the required refrigeration when tested as specified in 4.4.1. The control device shall be adjustable to maintain any desired temperature of 18°F to 23°F within the cabinet. Automatic refrigeration control shall be accessible for adjustment and servicing. Temperature adjustment shall be provided by means of a single adjusting knob. The "cut-out" position of the high pressure control shall provide safe operation by shutting down the compressor when a preset pressure reading is attained.

3.4.6.4 Condensate disposal. A condensate evaporator pan with the thermostatically controlled electric heater, and a drip pan drain pipe from the cooler drip pan direct to the evaporator pan shall be provided for disposal of condensate. Openings in the cabinet walls for passage of the drip pan drain pipe shall be mechanically sealed. The evaporator pan shall be removable and accessible for cleaning. The evaporator pan shall be located on top of the cabinet beneath the condensing unit or shall be shaped as a trough and mounted on back of the cabinet.

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3.4.6.5 Tube. All refrigerant tube shall be as specified in 3.3.1. The tube shall be arranged not to interfere with the servicing or removal of the condensing unit. Fittings shall be of wrought or forged copper. The refrigeration unit design shall prevent condensation on suction lines within the machine compartment. The run of suction line exposed to room air including the section in the machine compartment shall be insulated. Linear and butt ends and terminal ends shall be sealed with adhesive manufactured for that purpose by the same manufacturer supplying the insulation. Where liquid and suction lines are fastened together, the insulation shall be installed to cover both lines. Tube shall be installed with bends of ample radius to maintain full inside diameter and shall be clamped to rigid members or otherwise supported to eliminate vibration. A loop in suction line near the compressor shall be provided to minimize vibration. The loop shall be installed with sufficient clearance to prevent abrasion when refrigeration unit is operating. Where tubes of different materials are joined, a shrink-fit plastic sleeve or other method shall be used to prevent wetting of the surfaces for at least 1-inch to each side of the joint.

3.4.7 Defrosting. Defrosting shall be accomplished automatically with automatic reset and shall not require the removal of shakes during the defrosting period.

3.4.8 Electrical characteristics. The shake cabinet shall be furnished completely wired and ready for connection to a source of electrical power. The shake cabinet shall be designed for operation on a 120 volt, 60 Hertz (Hz), 1-phase system. The shake cabinet shall be provided with a three wire self-grounding supply cord a minimum of 6 feet in length with a three wire self-grounding type attachment plug. The shake cabinet shall be provided with an interior light, automatically controlled by a switch activated by opening and closing of the door. Means shall be provided for protection of wiring and receptacles and removal of the light.

3.4.9 Basket. The basket shall be constructed of heavy gauge chrome plated steel wire and measure 21 inches wide by 26-3/4 inches deep by 5-3/4 inches high. The basket shall hold a minimum of thirty-five (35) 16 fluid ounce cups measuring 3 inches in diameter at 2-3/4 inches high and 5-1/2 inches overall height. Eight (8) baskets shall be furnished with each shake cabinet.

3.4.10 Operation. Shake cabinet refrigeration system, controls, electrical components and moving parts shall perform as intended without binding, looseness or malfunction when tested as specified in 4.4.1.

3.4.11 Performance.

3.4.11.1 Shake cabinet, no load. The shake cabinet when no load tested as specified in 4.4.3 and 4.4.3.1 shall conform to the following requirements.

a. The refrigerant system shall be capable of lowering the empty internal temperature from 90°F ± 2 to 18°F in an ambient temperature of 90°F in not more than 6 hours.

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b. The compressor running time shall not exceed 80 percent.

c. The condensing unit motor shall not be damaged and shall not cut out on either inherent or external overload protectors.

3.4.11.2 Shake cabinet, load. The shake cabinet when load tested as specified in 4.4.3 and 4.4.3.2 shall conform to the following requirements.

a. The shake cabinet refrigeration system shall maintain an average milk shake temperature between 18°F and 23°F.

b. No single milk shake temperature shall exceed 28°F.

c. The compressor running time shall not exceed 90 percent.

d. The condensing unit motor shall not be damaged and shall not cut out on either inherent or external overload protectors.

3.4.12 Inclined operation. The shake cabinet shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product when tested as specified in 4.4.8.

3.4.13 Electromagnetic compatibility. When specified (see 6.2 and 6.4), the shake cabinet shall be designed and equipped for electromagnetic compatibility of Group I, class C3 of MIL-STD-461 and tested as specified in 4.4.6.

3.4.14 Fungus and moisture resistance. When specified (see 6.2), the shake cabinet shall be treated to resist fungus and moisture as follows:

3.4.14.1 Circuit elements. Circuit elements, which have a temperature rise of not more than 75°F when operating at full load, shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, terminal blocks, capacitors, and coils.

3.4.14.2 Electrical components not requiring fungus moisture resistance. Electrical components such as switches, fuses, and contacts, shall not be treated. Other materials and components which are inherently fungus-resistant or are protected by hermetic sealing need not be treated.

3.5 Marking. Each shake cabinet shall have a corrosion-resisting steel identification plate securely attached to the body of the unit. The plate shall include the national stock number (NSN), contract number, date of contract, manufacturer's name or trademark and model and serial numbers. Markings shall be permanent and legible.

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3.6 Finish. The shake cabinet shall be smooth on all surfaces and edges and shall be free from discoloration or stains. The components shall be finished in accordance with the manufacturer's standard practice.

3.7 Repair and maintenance. Provisions shall be made to allow ready adjustment, service, and replacement of electrical power system components and operational electrical accessories and components, without disturbing or disassembly of any major component or equipment, as required, for proper operation of the shake cabinet.

3.8 Workmanship. All components and assemblies of the shake cabinet shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, damaged, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners, except when corners are required.

3.8.1 Metal fabrication. Metal used in the fabrication of the shake cabinet shall be free from kinks. Forming and shearing shall not cause damage to the metal.

3.8.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

3.8.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Captive nuts, threaded fasteners shall be of the locking type and shall not be broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

3.9 Wiring. Wiring shall be enclosed, supported, and protected from grease or food spillage and against mechanical damage. Metal edges with which wire comes in contact shall be rounded and smooth. Wires shall be secured to prevent interference with servicing and operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Density of insulation. The density test for insulation shall be performed as specified in 4.4.7. The inspection lot shall be the amount of material used in one day's production. The sample unit for this test shall be a 6 inch cube of insulation.

4.2.2 In-process inspection examination. Examination shall be made of the following operations to establish conformance with specified requirements. Whenever nonconformance is noted, corrections shall be made to process and all items processed.

(a) Installation and supporting of piping (see 3.4).
(b) Accessories, instruments, controls, and tubing securely fastened or braced (see 3.4).
(c) Tapping plates installed (when provided) (see 3.4.1).
(d) Installation of insulation (see 3.4.1).
(e) Sealing of openings in cabinet shells (see 3.4.1.1).
(f) Requirements for outer shell and refrigerated compartment liner (see 3.4.1.1 and 3.4.1.2).
(g) Requirements for doors (see 3.4.2).
(h) Dehydration, evacuation, and charging of refrigeration system (see 3.4.6.3).
(i) Requirements for condensate disposal (see 3.4.6.6).
(j) Protection of wiring and receptacle (see 3.4.8).
(k) Circuit elements coated with varnish as required in 3.4.14.1.
(l) Welding (see 3.8.2).
(m) Captive nuts, threaded fasteners or other securing devices provided in restrictive areas or in access panel mounting areas (see 3.8.3).
(n) Wires secured to prevent interference with servicing and operation (see 3.9).

4.2.3 End item inspection. The lot shall consist of all shake cabinets offered for inspection at one time. The sample unit shall be one shake cabinet.

4.2.3.1 Visual examination. The completely assembled shake cabinet shall be examined for defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not manufacturer's standard product	X	
Finish	Edges and surfaces not smooth	X	
	Not free of discoloration or stains		X
Design, construction, and workmanship (general)	Part or component missing	X	
	Any rough or sharp edges, burrs, slivers or scratches; rough die or grind marks	X	
	Any component fractured, buckled, bent, punctured, malformed, or damaged	X	
	Component not readily accessible for servicing where required	X	
	Functioning component that does not operate as intended	X	
Cabinet	Not as specified	X	
	Component missing	X	
Gasket	Not replaceable	X	
	Not one piece	X	
	Not magnetic type	X	
Condensate disposal	Openings in cabinet wall for passage of drip pan pipe not sealed	X	
	Evaporator pan not mounted as specified		X
Doors	Miter joints not sealed	X	
	Miter joints not sealed by continuous welding	X	
	Not removable	X	
Door hardware	Not attached as specified	X	
Hinges	Not gravity action type	X	
	Not adjustable	X	
	Not equipped with stops		X
	Not provided with dwell	X	
	Not material specified	X	
	Not edge mounted type	X	
Latch and strike assembly	Not lockable	X	
	Not adjustable	X	

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TABLE I. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction, and workmanship (general) (cont'd)			
Door hardware (cont'd)			
Handle	Not provided	X	
	Not type specified		X
Door lock assembly	Not as specified		X
	Not provided	X	
Thermometer	Not digital type, temperature range not as specified	X	
	Not adjustable	X	
	Not provided with repositioning switch	X	
Supports	Not type specified	X	
	Not installed as specified	X	
	Not removable	X	
Legs	Not type specified	X	
	Not attached by bolts through gusset plates	X	
	Not provided with bottom flange with holes for hold down bolts	X	
Basket	Not quantity provided	X	
	Missing	X	
	Not as specified	X	
Condensing unit	No removable door or panel on compartment	X	
	Door or panels not louvered	X	
	Not accessible for servicing	X	
Condenser	Fins not soldered, welded, or mechanically bonded to tubes	X	
	Fins not spaced as specified		X
Motor starter	Not mounted in drip-proof enclosure	X	
Receiver	No fusible plug	X	

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TABLE I. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design, construction and workmanship (general) (cont'd)			
Cooling unit	Not mounted as specified	X	
Evaporator	Not finned-tube construction	X	
	Fins not spaced as specified	X	
Control system	Control device not adjustable to maintain temperature as specified	X	
	Temperature adjustment not as specified		X
Welds	Scale or flux deposits not removed; not ground or polished	X	
Sheet metal and structural sections	Kinks, or bends		X
Electrical wiring and connections	Cabinet not completely illuminated		X
	Switch for lights not operated by opening door		X
	Switch not plunger type		X
Identification markings	Missing, incomplete, not legible	X	
Markings (instruction and identification, as applicable)	Missing, incomplete, not legible	X	
	Not specified type or size, not affixed in prescribed manner		X

4.2.2.2 Dimensional examination. The shake cabinet shall be examined for compliance with dimensions specified. Any dimensions not within the specified tolerance shall constitute a defect. The inspection level shall be level S-2 with an AQL of 4.0 defects, expressed in defects per hundred units.

4.2.2.3 End item testing. One shake cabinet from the initial lot shall be tested as specified in 4.4.1 through 4.4.5; 4.4.7 and 4.4.9 and failure to pass all of these tests shall be cause for rejection of the lot. Acceptance of any subsequent lot shall be based on satisfactory completion of all the preceding tests conducted on the unit from the initial lot. Each shake cabinet shall be tested as specified in 4.4.1, 4.4.2, 4.4.4 and 4.4.5 and failure to pass all of these tests shall be cause for rejection of the individual cabinet.

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4.2.3 Packaging inspection. An examination shall be made to determine that preservation, packaging, packing and comply with the quality assurance provisions of MIL-R-12323.

4.3 Certificate examination. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Methods of inspection.

4.4.1 Operational test. Connect the completely assembled shake cabinet to an electrical power source having the same characteristics specified. To determine conformance to 3.4.6.3 and 3.4.10, the shake cabinet shall be operated at least 1 hour to demonstrate that all parts are in working order, that controls function according to settings, and that refrigeration system has been properly evacuated and dehydrated. The operating test shall be conducted at normal production room temperatures. Failure to comply with the requirements of 3.4.6.3 and 3.4.10 shall be cause for rejection of the item.

4.4.2 Door tightness test. To determine conformance to 3.4.2, the seating and the sealing of door gaskets shall be tested at the four corners and midpoints on all doors by trying to insert a 0.005 inch leaf-gage between the gasket and the seating face. Lack of noticeable resistance to the insertion of the gage shall be cause for rejection of the item. Any nonconformance with specified requirements shall be cause for rejection of the item.

4.4.3 Performance tests. The temperature sensing devices in the interior of the shake cabinet shall be located in each of the eight basket areas so as to provide data on average, high and low temperature of each basket area and for the entire interior area. Ambient temperatures shall be measured at points location 4 feet above the floor and 10 inches from the middle of each side and front of the cabinet. Ambient temperatures at each point shall be maintained within 90°F $\pm$ 2.

4.4.3.1 Shake cabinet, no load test. The shake cabinet shall be placed in an ambient temperature of 90°F with the door open and the cabinet electrically disconnected for eight hours prior to the start of the test. At the start of the test the cabinet shall remain at an ambient temperature of 90°F with the door closed and the cabinet electrically operated. Failure of the shake cabinet to meet the requirements of 3.4.11.1 shall constitute failure of this test.

4.4.3.2 Shake cabinet, load test. The shake cabinet shall be operated in an ambient temperature 90°F for at least 8 hours before the start of the test. During this stabilization period, the cabinet shall be fully loaded with eight baskets of milk shakes, (280 each of 16 fluid ounce cups), having product temperatures of not lower than 26°F. After the stabilization period, the test shall begin and be conducted for 8 hours. Failure of the cabinet to meet the no visible condensation requirements of 3.4.1 and the load requirements of 3.4.11.2 shall constitute failure of this test.

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4.4.4 Refrigerant leak test. Each shake cabinet shall be refrigerant leak tested to determine conformance with 3.4.6. The test for leakage shall be made with a halide leak detector or an electronic leak detector or any other method standard with the manufacturer. Any leakage shall constitute failure of this test.

4.4.5 Refrigerant moisture test. The refrigerant moisture and liquid indicator shall be checked for dryness as shown by the manufacturer's color chart data. A reading indicating moisture shall constitute failure of this test.

4.4.6 Electromagnetic compatibility test. When electromagnetic compatibility is required, one item shall be tested by the contractor in accordance with part 10 method UMO-5 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the EMI test plan, the electromagnetic test report and the end item meet MIL-STD-461 (see 3.4.13).

4.4.7 Density test for insulation. Prior to each day's production, one free foam sample of the insulation referenced in 3.3.3.1, a minimum 8 inch cube, shall be taken. From the center of the sample a 6 inch cube shall be cut and weighed. This weight shall be multiplied by 8 to determine conformance with the density requirement of 3.3.3.1. In addition, a daily record of the foam insulation shall be kept as follows and the production unit shall not deviate more than plus or minus 5 percent from the data recorded.

Calibration of mix (mix proportions)

Amount of flow (per box)

Time of flow (per box)

Mould temperature (per box)

Inability of sample to conform with density requirement of 3.3.3.1, or more than 5 percent deviation of data gathered on the production unit shall constitute failure of this test.

4.4.8 Inclined operational test. After the test in 4.4.3 is completed, position the shake cabinet with the base set at an angle of 15 degrees, then operate shake cabinet for 15 minutes each, at each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions observe for conformance with specified requirements of 3.4.12. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.9 Shelf load test. Place all angle supports in the normal stowed position in the refrigerated compartment. Load the basket with an evenly distributed load of 50 pounds and place the loaded basket on the angle supports for 10 minutes. The load removed, the angle supports shall be examined for compliance with the requirements specified in 3.4.3. Any noncompliance with the specified requirements shall be cause for rejection of the item. At each of the remaining seven positions the same test shall be performed for compliance with the requirements specified in 3.4.3. Any noncompliance with the specified requirements shall be cause for rejection of the item.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Each complete shake cabinet shall be cleaned, preserved, packed and marked in accordance with the applicable requirements of MIL-R-12323. Preservation and packing shall be level A or C, and packing shall be level A, B or C (see 6.2).

6. NOTES

6.1 Intended use. The shake cabinet is intended to be used in galleys for storing milk shakes prior to serving.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) What side door is to be hinged (see 3.4.2).
(c) When electromagnetic compatibility is required (see 3.4.13).
(d) When fungus and moisture resistance is required (see 3.4.14).
(e) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

6.4 Critical or non-critical areas.

6.4.1 Critical area. A location on a platform or installation containing equipment or subsystems which, if malfunctioning due to unwanted electromagnetic energy, could degrade the overall system performance and result in the failure or abortion of a primary mission. All locations on a submarine and surface ship are considered critical areas.

6.4.2 Non-critical area. A location in a ground installation where EMI will not result in failure of abortion of a mission or degradation of the overall performance. Examples of areas which may be considered non-critical are: office buildings, recreation areas, laundry areas, food servicing areas, drafting rooms and woodworking shops.

Custodians:

Army - GL
Navy - SA

Preparing activity:

Army - GL

Project No. 7320-0696

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-44044; Cabinet, Storage, Milk Shake, Mechanically Refrigerated

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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